

Assessing the Impact of COVID-19 on Inflation Dynamics in India: A Machine Learning Approach

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Abstract— This research work aims to study COVID-19's impact on the Indian economic system. The Consumer Price Index of India for different commodities from January 2013 to March 2023 is used for the dataset. The paper implements modern tools like Deep Learning (DL) and Machine Learning (ML) methods to analyse the impact COVID-19 had on the inflation rate in the economic state of India. This paper uses Linear regression, K-nearest neighbors, random RandomForest Regression, Gradient Boosting Regressor, and one deep learning LSTM model to analyse the Consumer Price Index dataset and the Unemployment Rate dataset from multiple open-source databases. During the study, comprehensive evaluation was done to calculate the model's performance with parameters like $r2_score$, Mean Square Error (MSE), Root Mean Square Error (RMSE), and (MAE) mean absolute error to forecast the CPI, and Inflation rate using the best performing model which was found to be the RandomForest Regression with the $r2_score$ of 0.990665. : We could see a sharp increase in both the inflation rate and consumer price index in 2020 at around 6.62% as expected and by using the best-performing model with an $r2_score$ of 0.99 the forecasting is visualized. This paper analyses and forecasts the inflation rate and CPI of India by implementing Machine learning methods adding new information to the already existing COVID data analysis.

Keywords: COVID-19, Inflation, Machine Learning, Deep Learning, Consumer Price Index.

1. INTRODUCTION

The global upheaval with the advent of the pandemic of COVID-19 has led to profound transformations in the economic landscape, compelling researchers to employ innovative methodologies to comprehend its multifaceted implications [1]. In this paper, a machine-learning approach is employed to analyse COVID-19's impact on the inflation rate in India [2]. India, with its intricate economic dynamics, experienced significant disruptions across different economic sectors, and as the nation strives to recover from the fallout, it becomes paramount to understand the specific dynamics of inflation [3]. In response to this demand, machine learning offers a transformative tool, capable of discerning patterns in vast datasets that include the Consumer Price Index, and

Unemployment rates across different sectors and providing nuanced insights into the evolving economic landscape [4]. This paper also delves into predictive measures of inflation rates to contribute while navigating the post-pandemic economic landscape.

The analysis and the prediction of the dataset were done using Linear Regression, Polynomial Regression, K-Neighbours Regressor, Random Forest Regressor, Gradient

Boosting Regressor, and one deep learning model LSTM model. These selected models ensure reliable outcomes that concern this pandemic issue, also the various individual factors can be identified and proper decisions can be taken likewise [2]. The existing body of literature on the impact COVID-19 had on the economic landscape reflects a dynamic and evolving field of research [1]. Most studies have delved into different factors that contributed to understanding ever changing economic landscape [1], but very less research have address the issue of Inflation during the COVID times using a machine learning methods which is the main motivation and the novelty of this paper. Inflation is a monetary phenomenon, When the money supply expands more rapidly than the growth rate of national income, inflation may ensue. Conversely, if the money supply and inflation escalate at an equivalent pace, inflation is unlikely to occur [5]. The stability of an economy can be known with inflation indicators that we can control at a moderate level. However, inflation is also needed for the rate of economic growth [6]. The main reason for inflation in India is due to inflation in food prices and other small factors [7]. It is very important to address this issue using a more efficient and effective modern technique instead of relying on the traditional purely statistical analysis, this proposed method of using a Machine Learning model will increase the understanding of the issue in a more accurate manner and can be easily improved in the future. In the subsequent sections, the paper delves into the machine learning methodologies employed, the selection and preprocessing of data, and the evaluation metrics guiding this

study provides a summary of the paper along with conclusions supported by references.

2. METHODOLOGY

2.1 Data Collection

In this paper, the overall details of the Consumer Price Index of India for different commodities from January 2013 to March 2023, and details of the Unemployment rate of India for different sectors from May 2019 to June 2020 were utilized, the dataset was taken from open-source resources like Kaggle, WorldBank.org, and Macrotrends. The datasets are organized and arranged chronologically with CPI for different commodities, different sectors, estimated unemployment rate, and labor participation rate. This data can be easily collected from the mentioned sites as it an open source data.

2.2 Data Preparation

The collected data is further pre-processed by eliminating null values, removing features that are irrelevant to the analysis, and adding important features such as the Inflation rate which is calculated from the CPI. The datasets are divided into 80% of data for training which is used to train the model and 20% of data for testing for evaluating the model's performance.

2.3 Workflow Diagram

The workflow diagram of the process is shown in Figure 1. From the figure, the entire workflow is subdivided into smaller partitions. The different sections are used for data visualization, data preparation, model training and evaluation, and data predictions. The model is trained with five different models i.e. Linear Regression, K-Nearest Regression, Gradient Boosting Regressor, Random Forest Regression, and Long Short-Term Memory (LSTM). The models' performance is calculated using different metrics like MSE, MAE, RMSE, and R2_SCORE parameters.

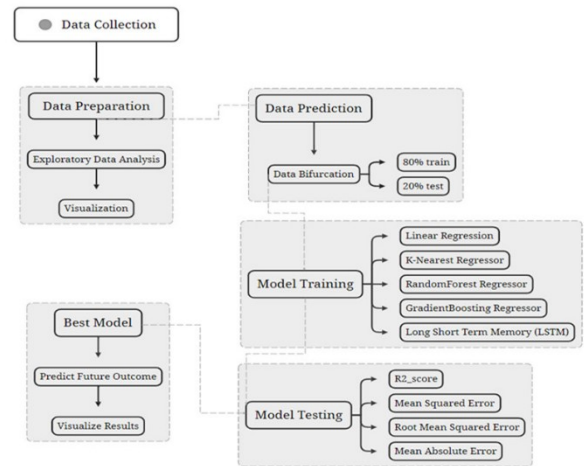


Figure 1: Workflow diagram.

The steps followed by the process are:

1. Initially, the datasets are collected from various reliable sources.
2. The datasets are then cleaned by removing the irrelevant features and handling the null values.
3. The dataset is analyzed and visualized using various Exploratory data analysis (EDA) techniques.
4. After the EDA, the dataset goes under data bifurcation where it is split into 80% of data for training and 20% of data for testing.
5. The training data is fitted on 5 different models, and it is evaluated using various scoring parameters on the testing data.

The best-performing model is noted and further utilized for forecasting and the results are visualized.

2.4 Data Visualization

In this Data Visualization section, the dataset was explored, and attempts to analyze the various patterns and relationships each feature has with the other features. The findings are further studied to develop a better understanding of the core dataset and important insights are drawn. In the following sub-sections, visualization of each aspect of the dataset and analysis of the nature of the data to draw valuable insights were performed. The role of EDA is important as we

can visualize the data and easily uncover anomalies, trends, test hypotheses, etc.

2.4.1 Sector Analysis

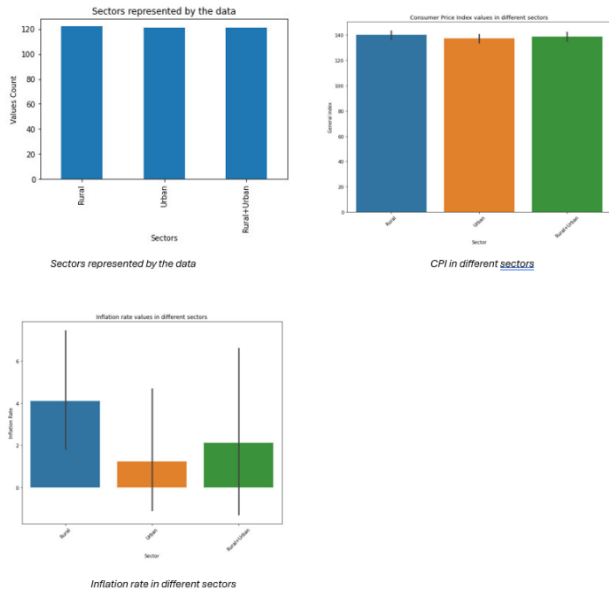


Figure 2: Sector Analysis

In this sector analysis, Figure 2 shows the data represented by different sectors, from this, it has been inferred that both urban and rural sectors are equally represented. It also shows the CPI and Inflation rate across the different sectors, and it has been inferred that the rural sector is more affected by inflation than the urban sector while the consumer price index is almost the same for both sectors. The visualization of the total CPI and Inflation rate for both sectors was also carried out. This analysis can be achieved by making bar graphs of rural, urban, and a combination of both for comparison.

2.4.2 Analysis of Consumer Price Index and Inflation Rate

In this CPI and Inflation Rate analysis, the inflation rate can be calculated using the Consumer Price Index i.e.

$$\text{Inflation rate} = \frac{CPI_2 - CPI_1}{CPI_1}$$

Where CPI_2 is the CPI in the second period and CPI_1 is the CPI in the previous period. As the Inflation rate is directly dependent on the CPI, it is a good measure to calculate our Inflation rate from it.

From Figure 4. Visualization of the inflation rate and the CPI over a decade is done and inferred that in 2020, the inflation rate spiked to a maximum during the COVID-19 period.

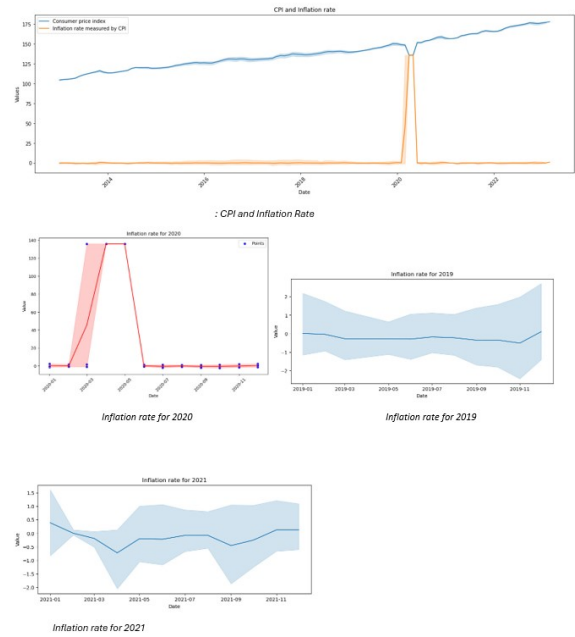


Figure 3: CPI and Inflation Rate Analysis

From Figure 3, a deeper analysis of 2020 statistics is studied and compared with pre and post-COVID years as shown in Figures. In 2020, the inflation was most severe during the summer when the number of active cases also reached its maximum. In 2019 and 2021, the inflation rates have normalized once again.

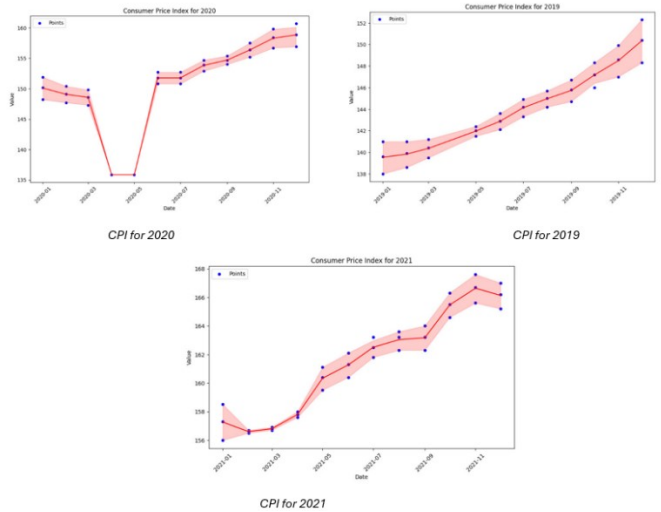


Figure 4: CPI for different years

In Figure 4, an analysis of 2020 statistics for the consumer price index is executed and compared with pre and post-COVID years. From Figure 6, it has been inferred that the CPI was at its lowest during the summer of 2020 when the

inflation rate skyrocketed. In 2019 and 2021, the CPI has a smoother curve once again.

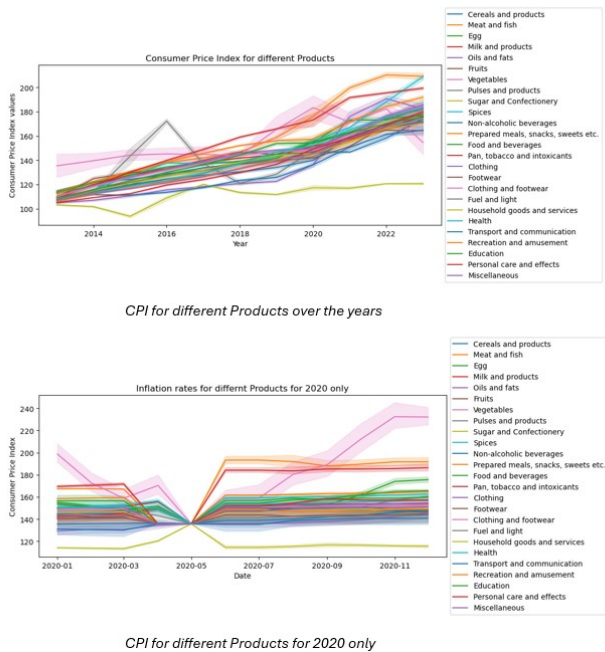


Figure 5: CPI and Inflation Rate for different products over the years

Figure 5 visualizes the CPIs for different commodities in India, after analyzing the different commodities, the sharp spikes in CPI for necessary commodities in 2020 are noted. Studying in further detail 2020's CPI only, it has been inferred that all the different CPI for different commodities reached a common value at around May 2020.

3.4.3 Unemployment Rate Analysis

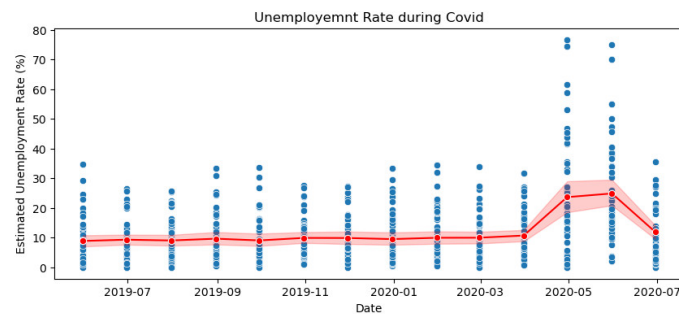


Figure 6: Unemployment Rate during Covid

In the Unemployment rate analysis in India, from Figure 6, it has been inferred that the unemployment rate in India began rising in March of 2020 and reached its peak in May of 2020 when heavy lockdown was imposed by the government and

over the following months the unemployment rate has been slowly coming down.

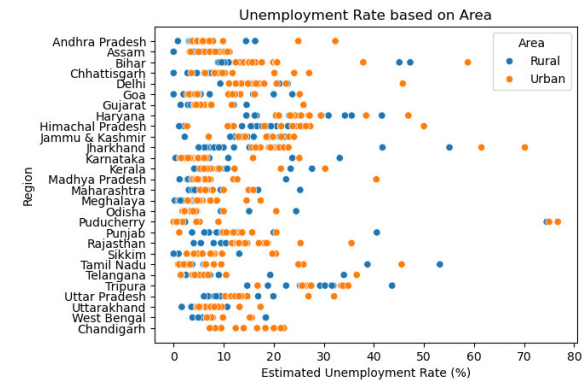


Figure 7: Unemployment Rate based on area

In Figure 7, a visualization of the different states that have been affected is conducted and analyzed based on the different sectors of the states. At 76% of the unemployment rate, Punjab suffered the highest out of all the different states, and Urban Punjab was slightly more affected by unemployment than rural Punjab. It is noted that small union territories like Puducherry have been the least affected out of all the different areas.

2.4.4 Comparative Evaluation

In this section, the analysis has been done comparatively according to the visualizations in the above sections. According to the statistical analysis performed, the subsequent key observations were noted.

1. All the anomalies in the data were found to be during COVID-19, therefore COVID-19 had a lot of impact on different areas of the economic landscape of India.
2. As the inflation rate during COVID-19 rises, it reached its maximum peak during the summer of 2020, strict government policies and heavy lockdown procedures have contributed to this cause.
3. The consumer price index of different commodities is studied, and it is found that out of all the commodities, the vegetable price shot up during 2020 and it is also noted that all the different CPI reached a common point in 2020.
4. The Unemployment rate has been compared with the inflation rate and it has been inferred that the spike between the inflation rate and unemployment rate occurred during the same period and its relationship with each other has been established. As the unemployment rate increased it has been found that the inflation rate also increased.

2.5 Data Prediction

2.5.1 Data Bifurcation

The entire dataset is split into two parts. 20% is reserved for testing and evaluating the model performance and the remaining 80% is used for training out data with different models for utmost efficiency.

2.5.2 Regression Models

In this paper, five different algorithms have been selected based on their usage and their different architectures, we train the data on all 5 of them and perform a comparative analysis to decide the best one.

1) Linear Regression Model

Linear regression is a statistical procedure that calculates the value of a dependent variable based on an independent variable. It assesses the link between two variables by utilizing a modeling approach to forecast a dependent variable from one or more independent variables [8]. This approach is frequently used for both predictive modeling and studying the relationships between variables [9]. The formula used to calculate linear regression is:

$$y = b_0 + b_1x \quad (1)$$

where b_0 is a constant value, b_1 is also known as regression coefficient or the slope; x denotes the independent

variable while y signifies the anticipated value of the dependent variable^[10].

2) K-Nearest Neighbours Regression

The k-Nearest Neighbors (KNN) algorithm is fairly adaptable and intuitive in machine studying, serving functions in both classification and regression duties. Throughout the schooling phase, the set of rules memorizes all training times that is used to create a reference dataset. When making predictions for new, unseen records, the algorithm evaluates the space to all the points within the schooling dataset, typically the use of a distance metric consisting of Euclidean distance^[11]. The Euclidean distance (d) between two points p and q in a 2-dimensional space is given by the formula^[12]:

$$d(p, q) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2} \quad (2)$$

For a general n-dimensional space, the formula is:

$$d(p, q) = \sqrt{\sum_{i=0}^n (p_i - q_i)^2} \quad (3)$$

The nearest k-neighbors to another data point are determined by calculating these distances. In the classification task, the algorithm assigns the most common class label to the k nearest neighbors by using a voting algorithm. In the regression function, the algorithm computes the average (or weighted average) of the target values associated with the k-nearest neighbors. The choice of distance measure is important for fine-tuning the model and depends on the nature of the data, in general, Euclidean distance is the most commonly chosen K value represents the number of neighbors to consider, this is another parameter that must be defined first before the model is trained. Although KNN is conceptually simple, its performance for large data can be a concern and the value of K and the choice of an appropriate distance metric significantly affects its performance^[13].

3) Random Forest Regression

The Random Forest regression algorithm is an ensemble learning method that constructs a collection of decision trees during training and combines their predictions for robust results. In the training phase, a forest of decision trees is built, with the number of trees as a specified hyperparameter. Every tree is trained on a randomized subset of features, introducing diversity, and is also trained on a random subset of the training data with replacement, a technique known as bootstrap sampling. The prediction phase involves passing a new data point through each decision tree in the forest to obtain individual predictions. For regression tasks, the final prediction is often the mean of these individual tree predictions^[14].

The Random Forest regression formula can be expressed as follows: Given a new input data point X , let

$\{y_1, y_2, \dots, y_n\}$ be the predicted outputs from each tree in the forest. The final prediction y_{final} is the mean of these predictions^[15]:

$$y_{final} = \frac{1}{n} \sum_{i=1}^n y_i \quad (4)$$

4) Gradient Boosting Regression

Gradient Boosting Regression is based on a new learning technique called ensemble that sequentially construct a series of weak learners such as decision trees to correct the errors of the previous models^[16]. In the training phase, a weak learner, denoted as $F_0(x)$, is trained on the original data. The

residuals, representing the differences between the predicted values and actual values, are computed for each data point. Subsequent weak learners $h_1(x), h_2(x), \dots, h_n(x)$, are trained to predict these residuals, and each new learner is added to the model with a weight that minimizes the overall loss function.

The gradient-boosting regression formula for predicting the output $F(x)$ for a given input data point x is expressed as follows:

$$F(x) = F_0(x) + \eta \cdot h_1(x) + \eta \cdot h_n(x) \quad (5)$$

Here, η is the learning rate, determining the contribution of each weak learner to the final prediction. In the prediction phase, a new data point is passed through each weak learner in the sequence, and the final prediction is the sum of the predictions from all weak learners. Gradient Boosting excels in capturing intricate relationships within the data and mitigates overfitting by iteratively refining the model's predictions based on the residuals of previous iterations [17].

5) Long Short-Term Memory (LSTM)

The Long Short-Term Memory (LSTM) model is a recurrent neural network (RNN) architecture designed to effectively capture and model long-range dependencies in sequential data, making it particularly suitable for tasks involving time series and natural language processing. In the training phase, the LSTM's memory cells, and gating mechanisms are initialized, allowing the model to learn patterns and dependencies over

Table 1. Model Scores

Model Names	RMSE	MSE	MAE	R2_SCORE
Linear Regression	3.260529	10.631050	2.208699	0.970353
K-Nearest Neighbors	3.152178	9.936229	2.637651	0.970695
Random Forest Regression	1.863387	3.472210	1.471595	0.990665
Gradient Boosting	1.935275	3.745288	1.560110	0.989979

varying time scales. Training involves the sequential processing of data, with the model adjusting its parameters through backpropagation through time. The unique architecture of LSTM allows it to selectively retain or discard information, addressing challenges such as the vanishing gradient problem often encountered in traditional recurrent neural networks [18].

In the prediction phase, a new sequence or time series is passed through the trained LSTM, leveraging its recurrent nature to maintain an internal state capturing information from the entire input sequence. The final prediction is then generated based on the learned patterns and dependencies within the sequential data [19].

The formula for the forget gate in an LSTM cell is as follows:

$$f_t = \sigma_g(W_f x_t + U_f h_{t-1} + b_f) \quad (6)$$

The adaptability and effectiveness of the LSTM in capturing complex sequential relationships arise from the interplay of these equations during each time step [20].

2.5.3 Model Evaluation

In this section, the performance of each model is evaluated by calculating its RMSE, MSE, MAE, and r2_score [21]. Table 1 visualizes the r2_score for each model and the RadomForest Regressor is performing the best out of all 5 algorithms by scoring 0.99 in the r2_score evaluation.

Long Short-Term Memory	0.1214593	0.0003811	0.01475236	0.947950
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The above table shows the metrics for each model that have been employed and concluded that RandomForest Regression is the best-suited algorithm for the current dataset, and further utilization of this algorithm is implemented for future predictions of the inflation rate in India post-COVID-19. The model performance might be further enhanced using a much larger dataset or extremely fine-tuning the hyperparameters of the model.

3. RESULTS AND DISCUSSION

The investigation performs forecast on the consumer price index for a decade (2023-2030) and from the analysis of all the models' performance, RandomForest Regression outperformed all the models, Linear Regression and K-Nearest Neighbors performed the same and achieved nearly the same r_2 score. The deep learning model LSTM performs exceedingly terribly in this circumstance since neural networks tend to perform better in a much larger dataset. Be that as it may, the model forecasts that the CPI will increase in the coming years as shown in Table 1. To minimize the error distribution of our predictive model we have also implemented an ensemble gradient-boosting algorithm.

Using the CPI values, the Inflation rate is calculated and visualized in Figure 18, from this the Inflation rate has been forecasted for a decade and insights are drawn.

- There is a decrease in the inflation rate in the 2023 period as a part of normalization of post COVID-19
- The CPI keeps on increasing, which is an alarming sign for the economic landscape of the country.
- The policymakers should be kept aware of the sharp spikes in the inflation rate in the 2025 and 2028 period.
- The unemployment rate also decreases as the inflation rate decreases.

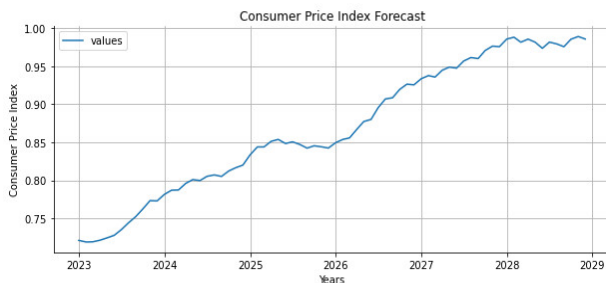


Figure 8. Consumer Price Index Forecast

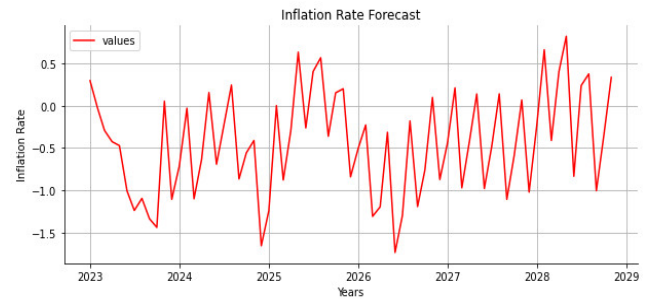


Figure 9. Inflation Rate Forecast

Finally, the paper conducts a thorough comparative analysis with previously published work, and the key points for takeaway of the study are:

- During the study, a deeper analysis of the inflation rate in India was conducted.
- The study visualizes the CPI of different commodities over the years.
- The work also highlights the effect of the inflation rate on the rate of unemployment in different sectors in India.
- Deeper Analysis in 2020 is carried out in the study and compared with the subsequent years.

Machine Learning models are utilized, and prediction of the CPI and the Inflation rate has also been done.

This paper provides a better understanding of the issue and the complex factors involved in the process. Unlike previous articles and reports written using traditional analysis, this paper gives the best understanding of the issue more qualitatively and also could be used to forecast the Inflation rate with 99% accuracy which can be used for further studies and analysis and paves the way for further research.

4. CONCLUSION

This study has introduced the Consumer Price Index from January 2013 to March 2023 as included in the study. The study proceeded with the analysis of the data and showed the rate of inflation and the unemployment rate in India are co-related. The study has shown the main cause of inflation is the rise in the consumer price index of common commodities with important items like vegetables reaching record high during the pandemic. In the study of the past 10 years in this paper, the highest inflation rate was observed in the summer of 2020, and

it was shown that it was the highest recorded in the decade. During the same period of the highest inflation rate, the highest unemployment rate was also observed as a result of the pandemic. The data further shows the different sectors affected by unemployment and inflation during the pandemic for different states. Punjab in particular was the most affected by this crisis reaching nearly 76% in unemployment rate. The machine learning models that were deployed in this study have provided valuable insights into predicting and forecasting the inflation rate for another decade. As for the study, prediction was done using five regression models to determine the best model, and based on that model the forecast of the rate of inflation and the consumer price index is accomplished. The experiments were concluded with the best MSE value as 3.472210, MAE as 1.471595, RMSE as 1.863387, and R2_score as 0.990665 using the Random Forest Regression model for the analysis of the Inflation rate.

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